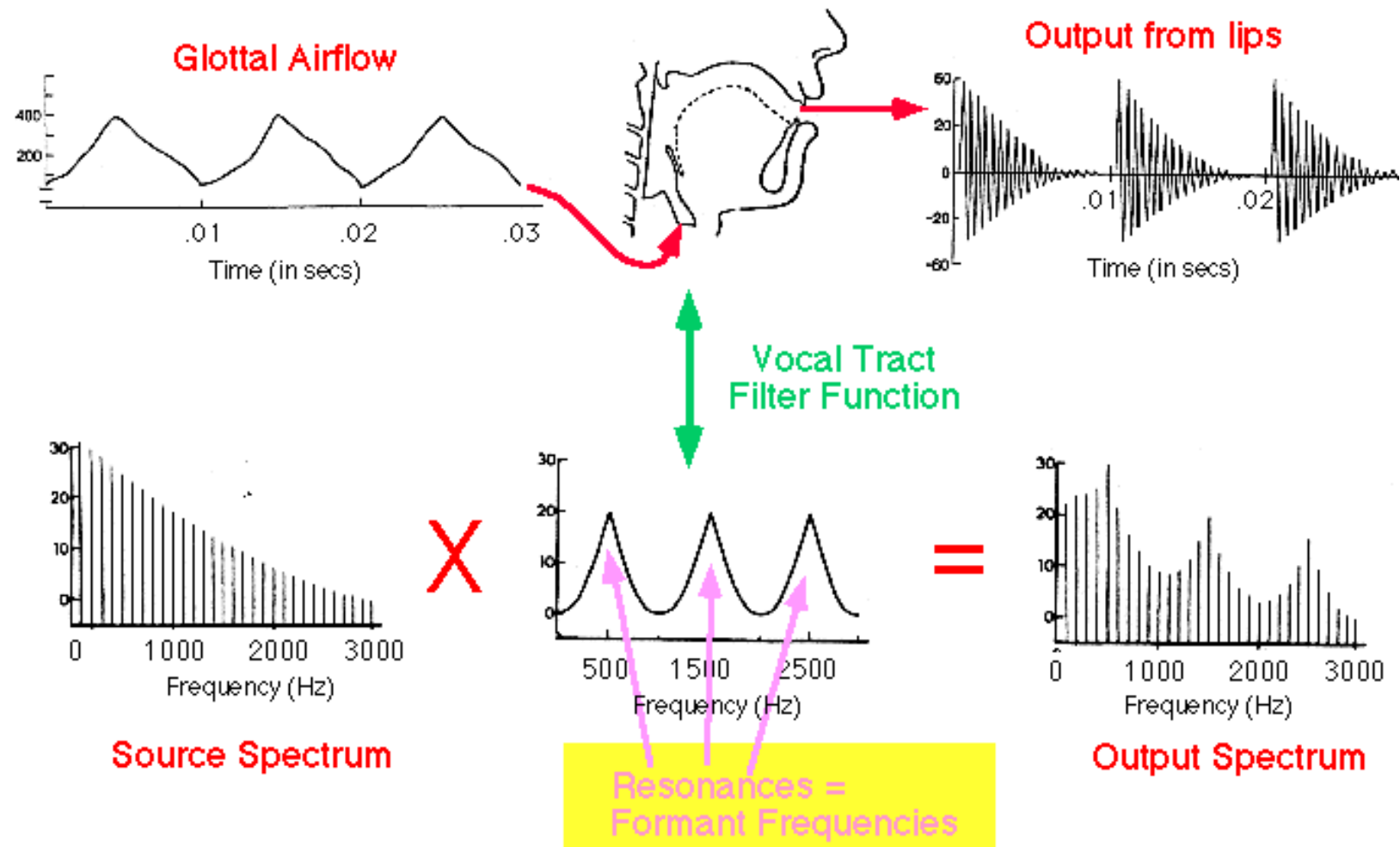


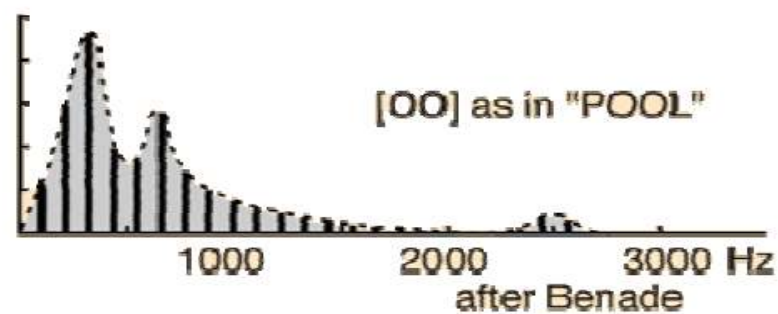
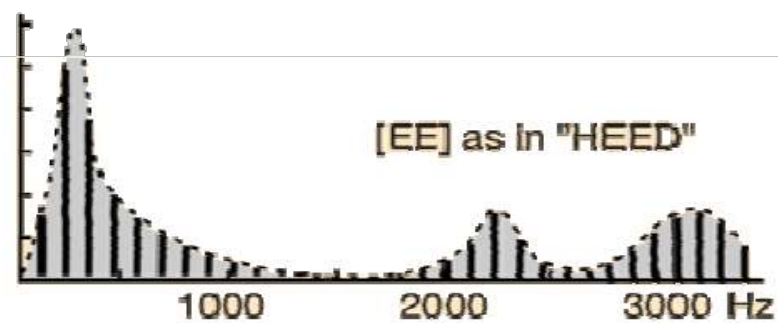
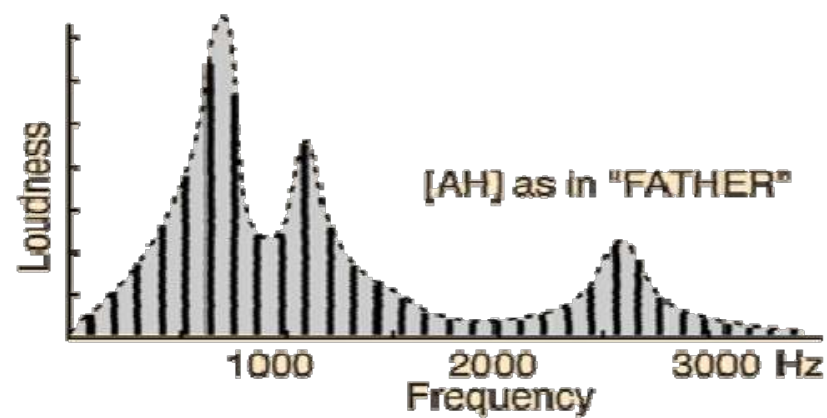
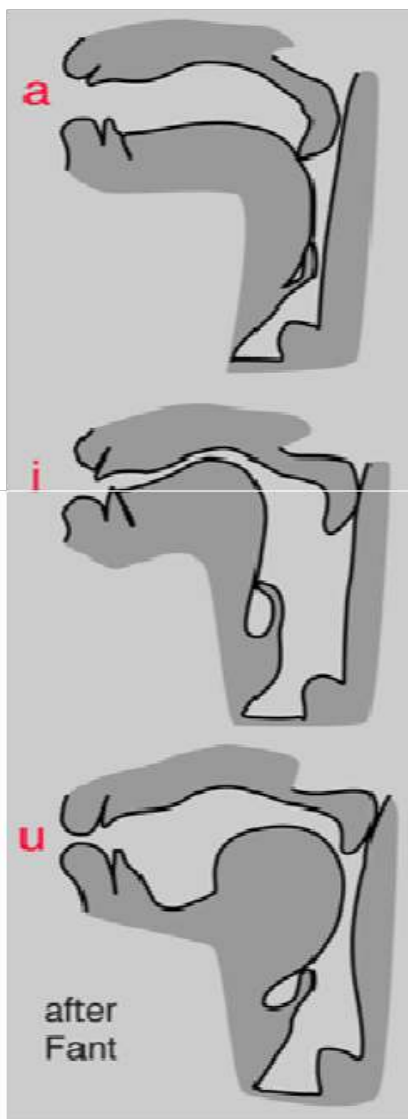
E9 205 Machine Learning for Signal Processing
21-8-17

Outline

- Melfrequency cepstral Coefficients (MFCC)
- Linear Prediction

Source filter model of speech

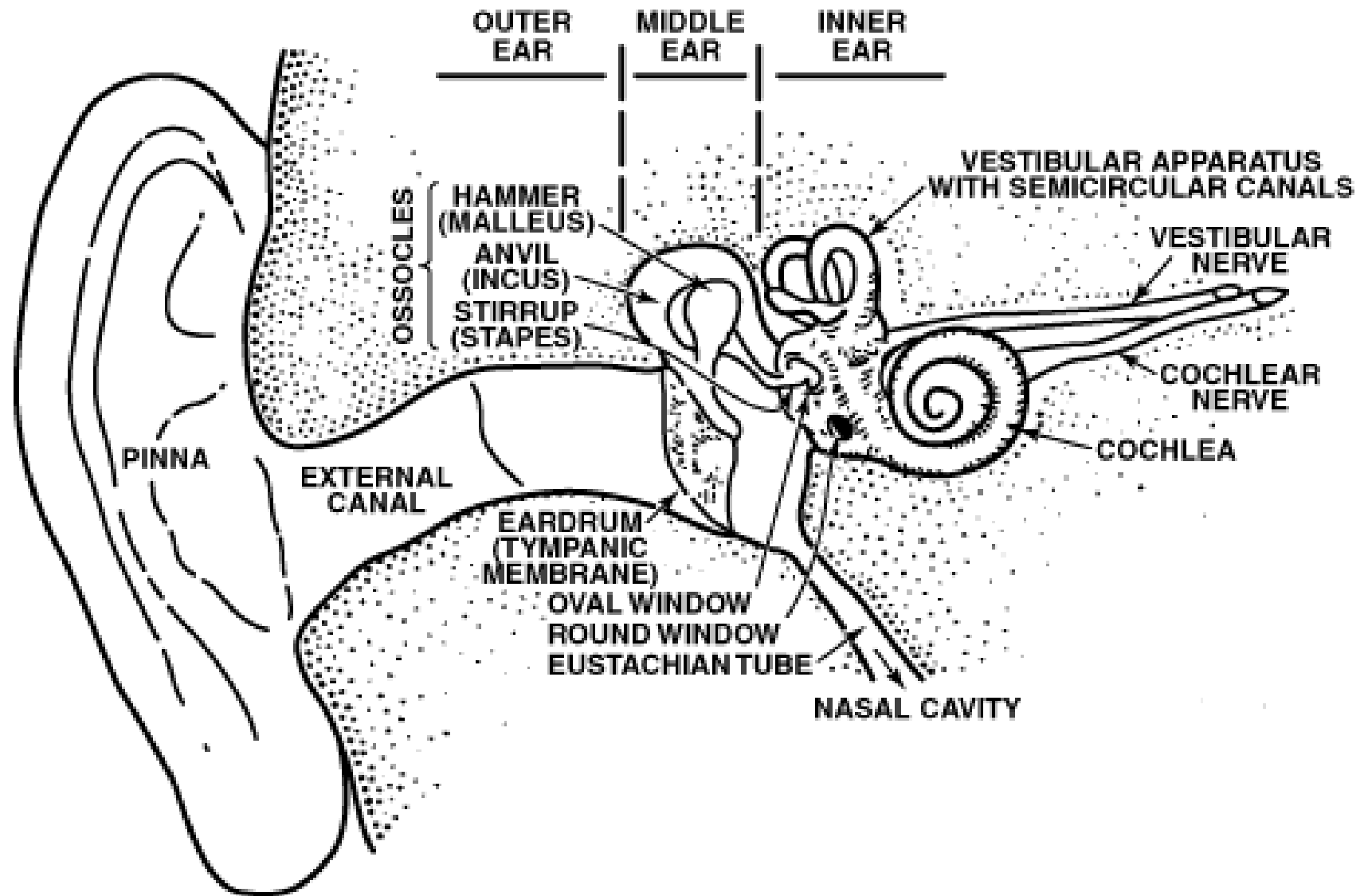




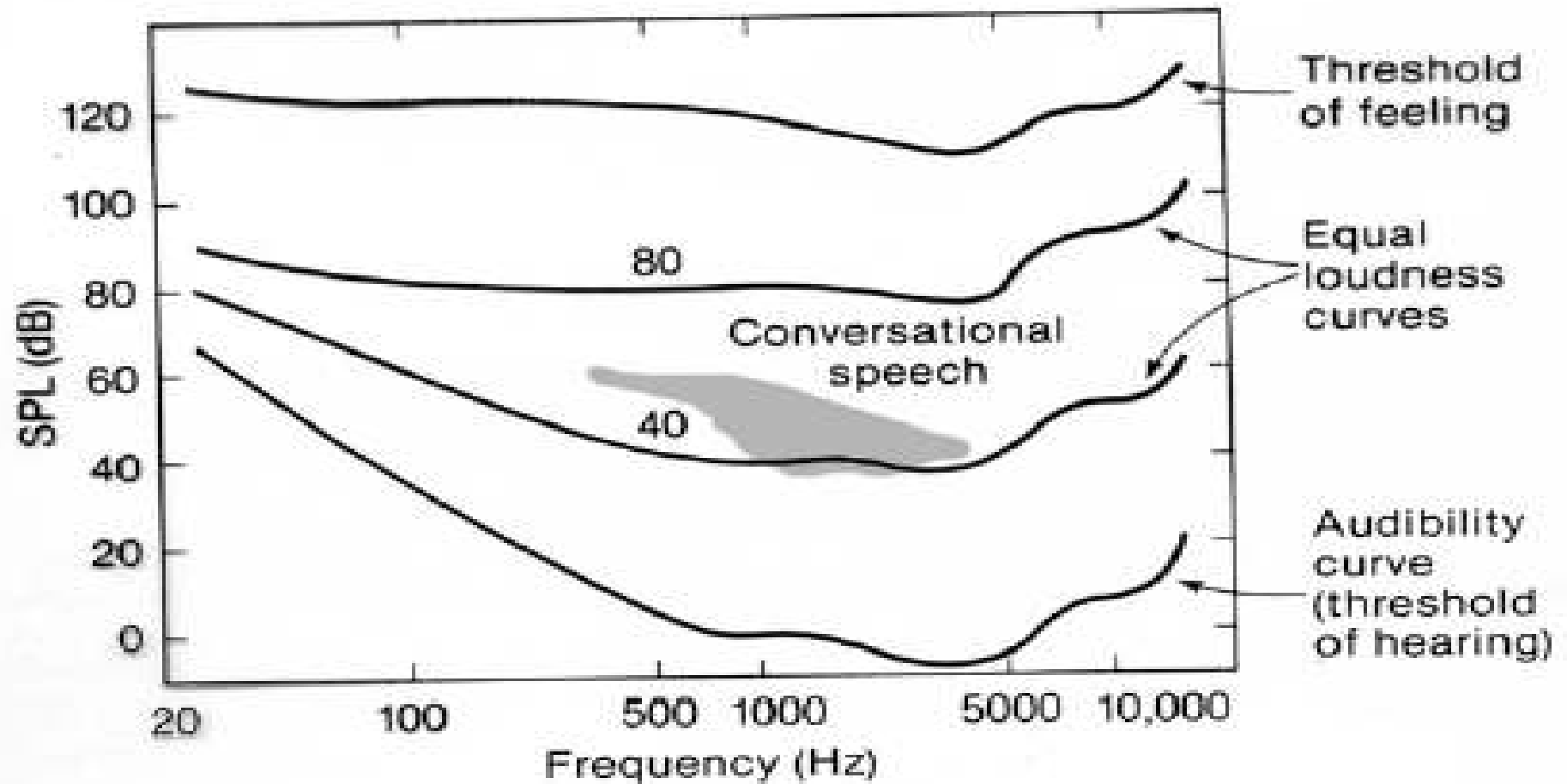
MFCC

- MFCC coefficients model the spectral energy Distribution in a perceptually meaningful way
- Why do we need?
 - Automatic speech recognition
 - Speaker Identification
 - Audio classification

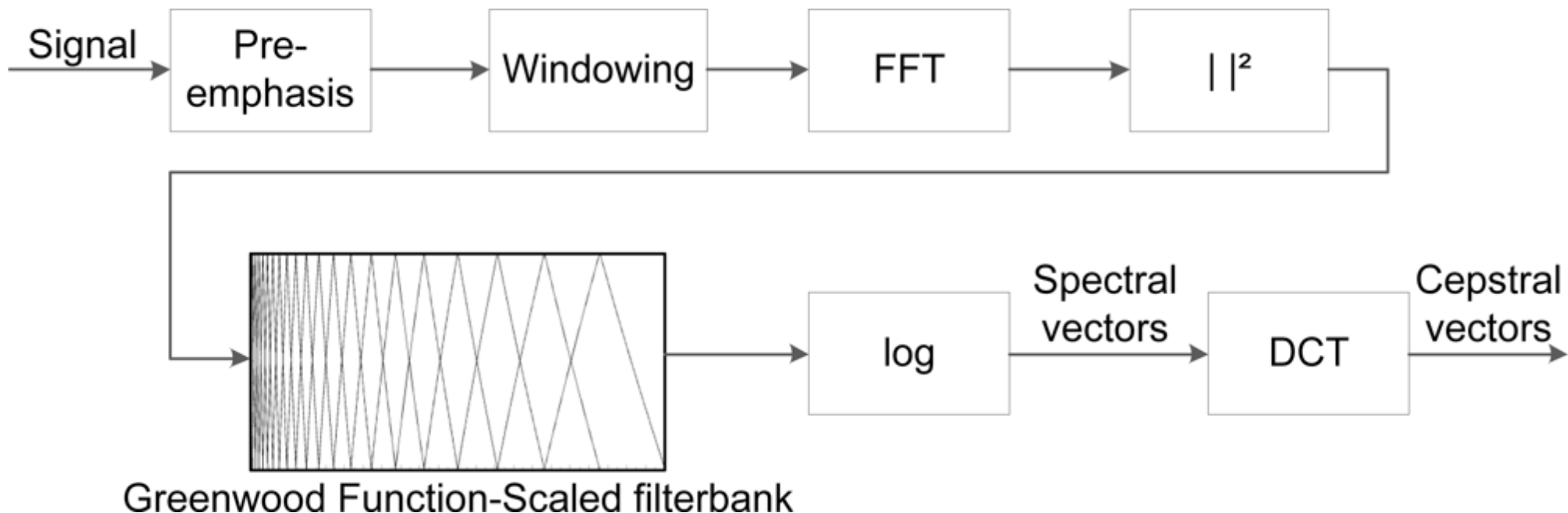
How do we hear?



- Equal loudness contours (loudness $\neq$ Intensity)



- Implementation steps



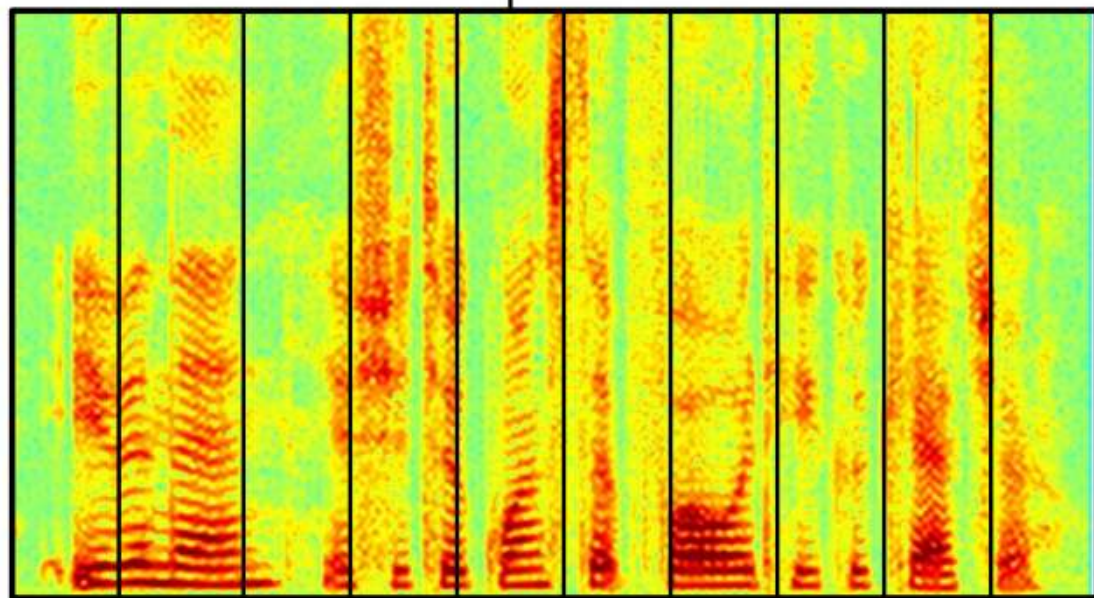
References:

[1]Chapter 4: Mohamed, Abdel-rahman. "Deep neural network acoustic models for asr." PhD diss., 2014.

[2]

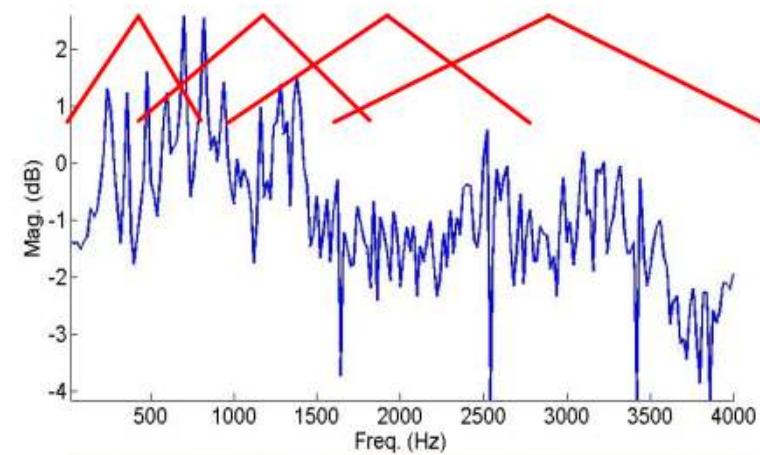
<http://practicalcryptography.com/miscellaneous/machine-learning/guide-mel-frequency-cepstral-coefficients-mfccs/>

Frequency

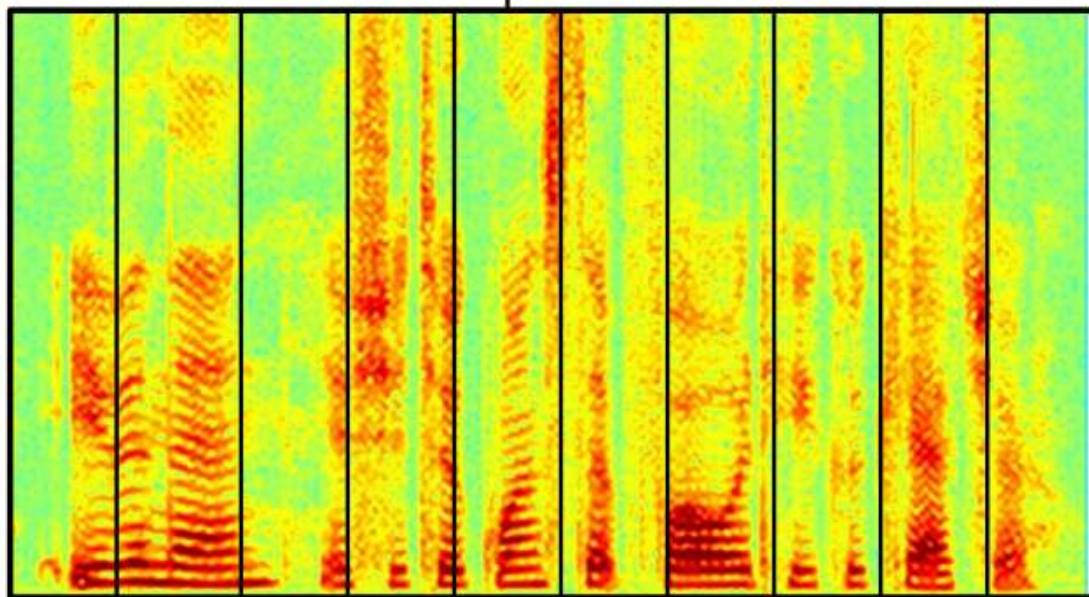


Time

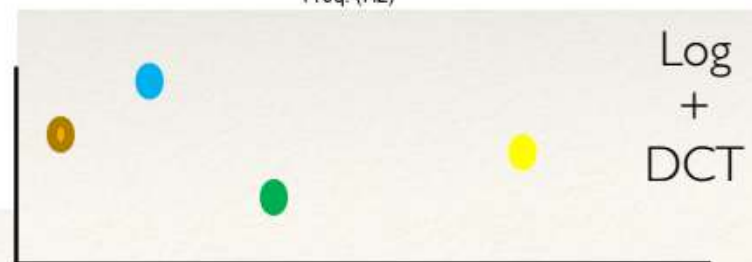
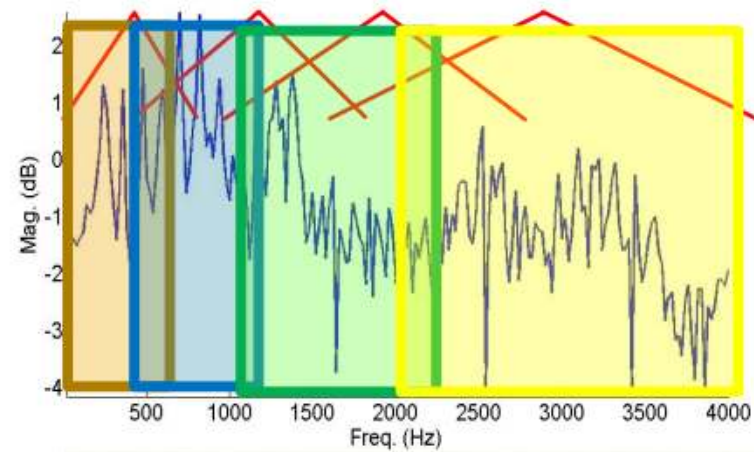
Mel



Frequency

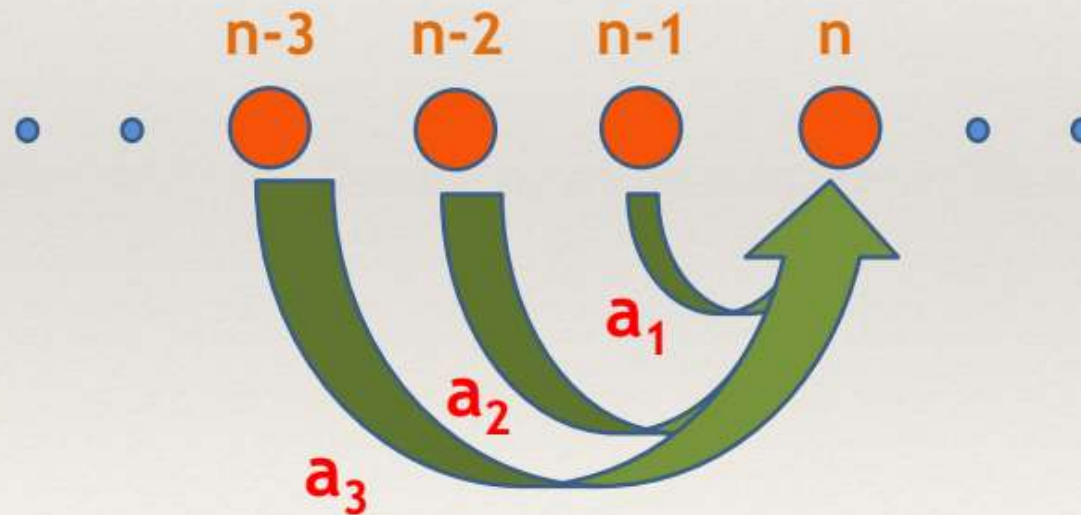


Time



Linear Prediction

- ❖ Current sample expressed as a linear combination of past samples



Properties of LP

Error signal (for the optimal predictor) is orthogonal to the samples used in the predictor.

$$e[n] \perp \{x[n-1], \dots, x[n-N]\}$$

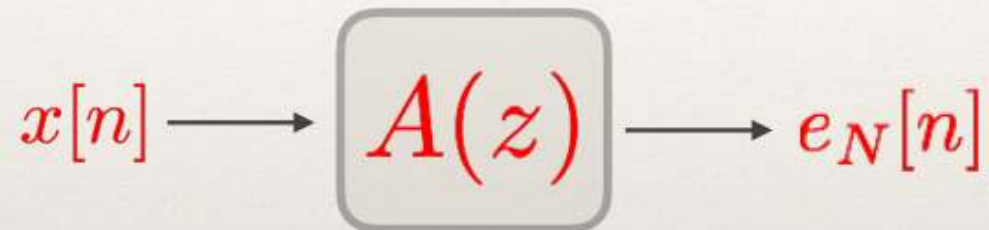
Using the orthogonality property $\rightarrow$ normal equations

$$\mathbf{R}\mathbf{a} = -\mathbf{r}$$

Autocorrelation matrix is Hermitian symmetric.

Properties of LP

Forward linear prediction filter



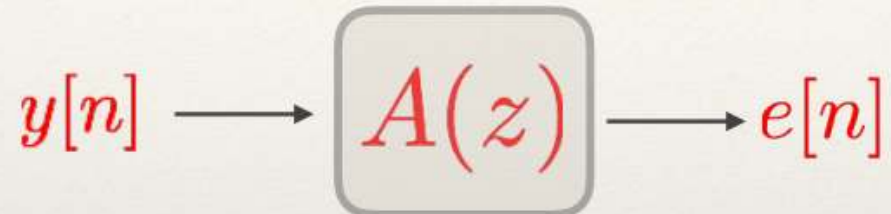
Properties of $A(z)$ - stability (all roots q)

$$|q| < 1$$

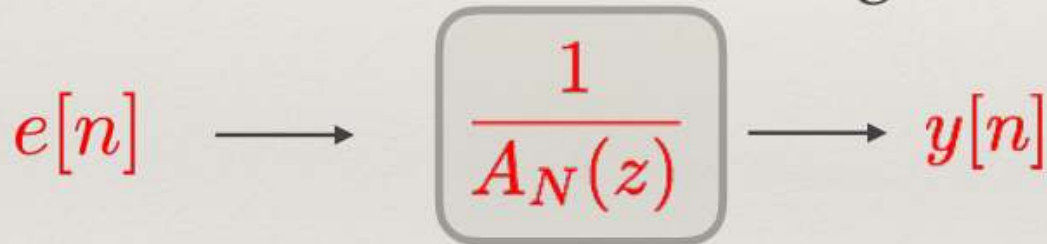
except for line spectral process $|R(k)| = R(0)$ for some k

Properties of LP

AR(N) process - Any WSS process which satisfies



Filter is stable - error signal is white



$$S_{yy}(f) = \frac{\epsilon_N}{|1 + \sum_{n=1}^N a_{N,n}^* e^{-j2\pi f n}|^2}$$

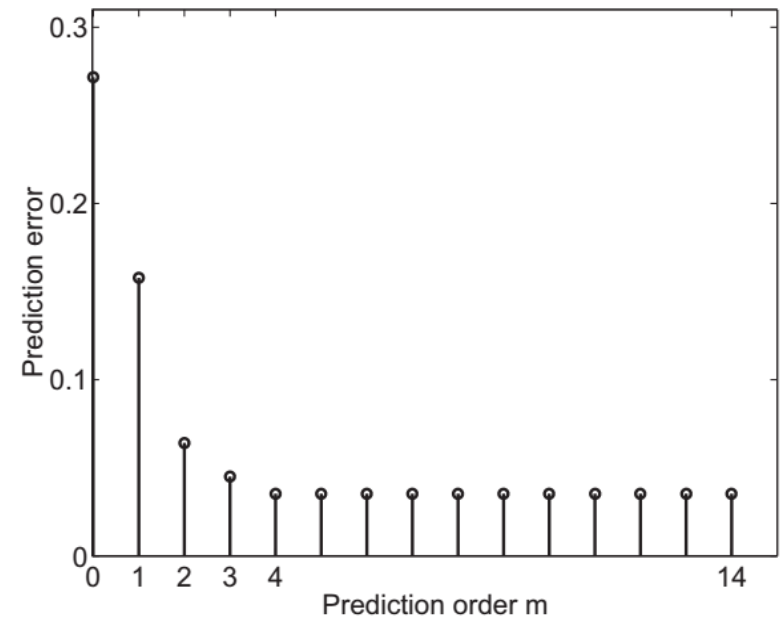
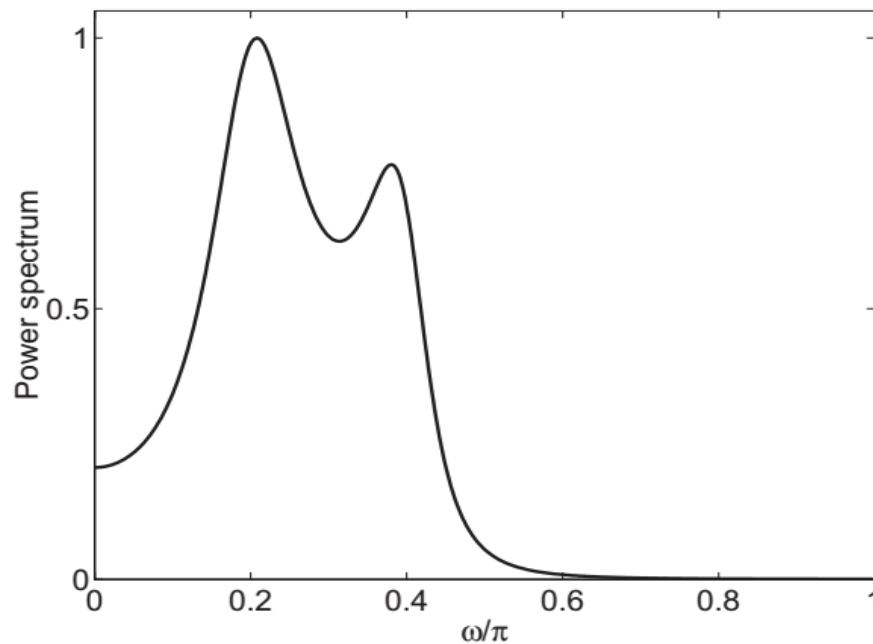
Approximating $x[n]$ by $y[n]$ i.e. $S_{xx}(f)$ with $S_{yy}(f)$

Autoregressive modeling

- AR(4) process:

- For AR process, error decreases monotonically and then becomes a constant for $m \geq 4$.

$A_0(z)$	1.0						
$A_1(z)$	1.0	-0.6473	0	0	0	0	0
$A_2(z)$	1.0	-1.1457	0.7701	0	0	0	0
$A_3(z)$	1.0	-1.5669	1.3967	-0.5469	0	0	0
$A_4(z)$	1.0	-1.8198	2.0425	-1.2714	0.4624	0	0
$A_5(z)$	1.0	-1.8198	2.0425	-1.2714	0.4624	0	0
$A_6(z)$	1.0	-1.8198	2.0425	-1.2714	0.4624	0	0



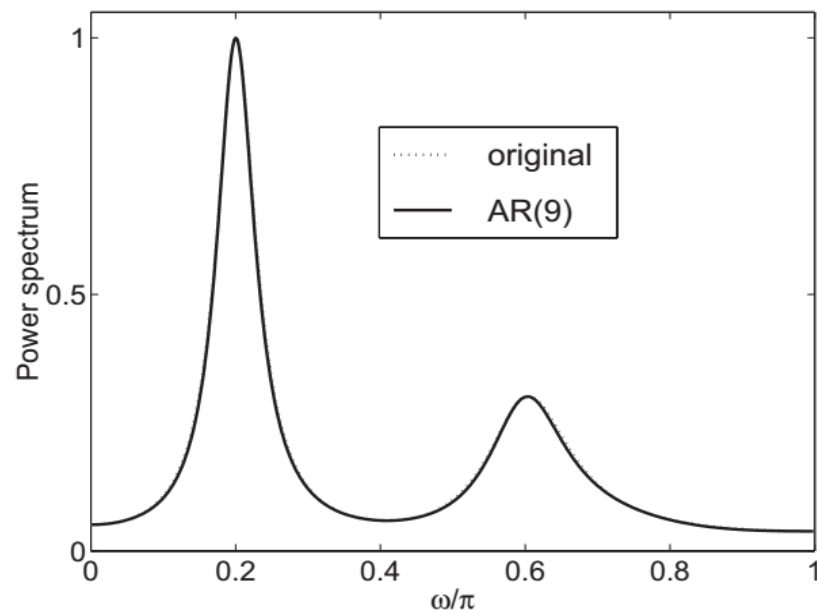
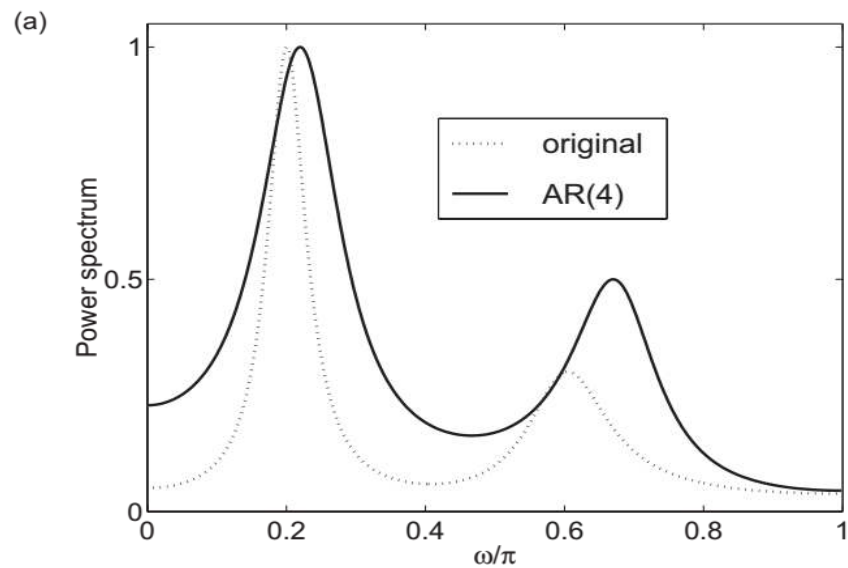
AUTOCORRELATION MATCHING PROPERTY

- Let $R(k)$ and $r(k)$ be the autocorrelations of $x(n)$ and the AR(N) approximation $y(n)$, respectively. Then,

$$R(k) = r(k), \quad |k| \leq N$$

Thus, the AR(N) model $y(n)$ is an approximation of $x(n)$ in the sense that the first $N + 1$ autocorrelation coefficients for the two processes are equal to each other.

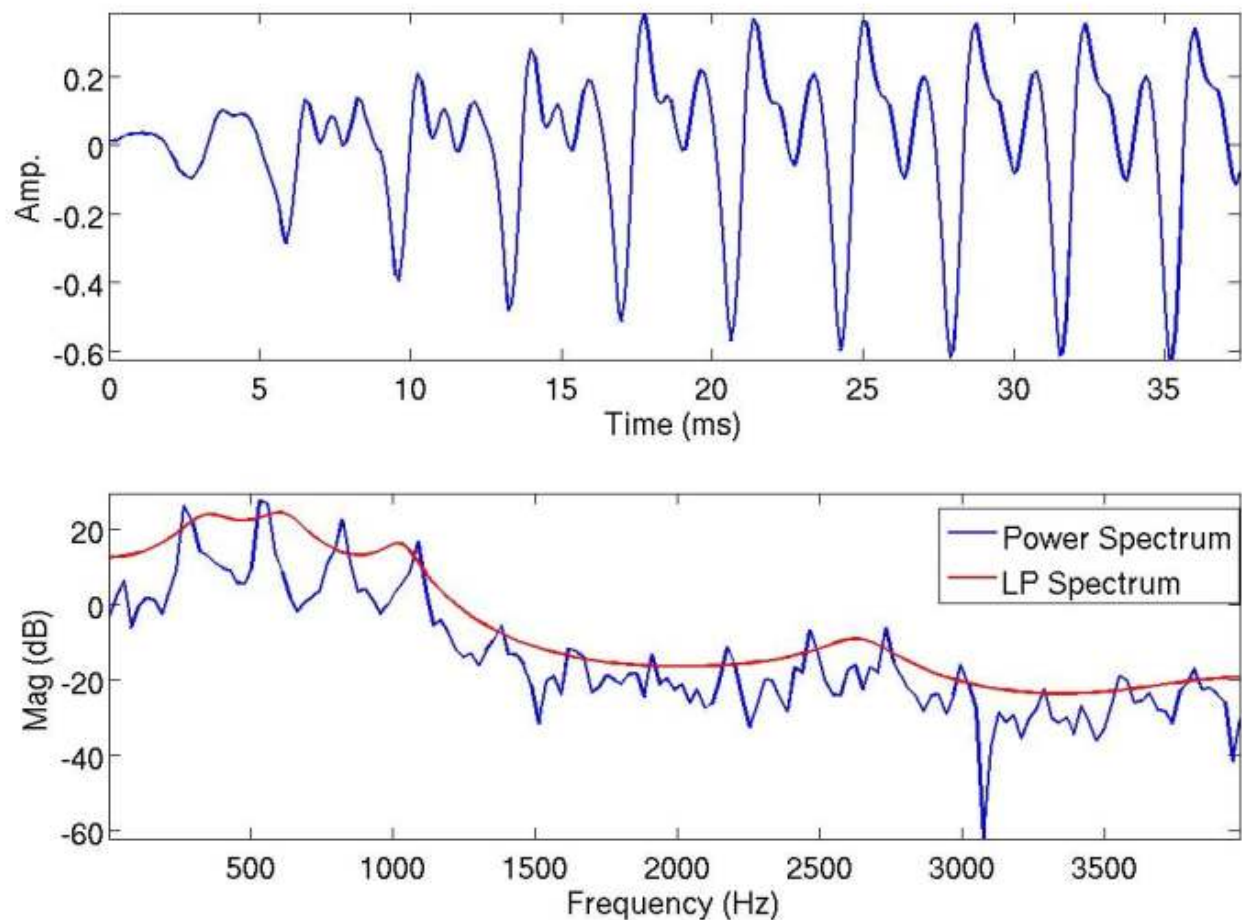
$$G_4(z) = \frac{3.9 - 2.7645z^{-1} + 1.4150z^{-2} - 0.5515z^{-3}}{1 - 0.9618z^{-1} + 0.7300z^{-2} - 0.5315z^{-3} + 0.5184z^{-4}}$$



m	$R(m)$	$r(m)$
0	0.1667	0.1667
1	0.0518	0.0518
2	-0.0054	-0.0054
3	0.0031	0.0031
4	-0.0519	-0.0519
5	-0.0819	-0.0819
6	-0.0364	-0.0364
7	-0.0045	-0.0045
8	0.0057	0.0057
9	0.0318	0.0318
10	0.0430	0.0441
11	0.0234	0.0241

Linear Prediction

AR Model of the Power Spectrum of the Signal



References

- [1] Vaidyanathan, P. P. "The theory of linear prediction." Synthesis lectures on signal processing 2.1 (2007): 1-184.
 - Chapter 2, 5
 - Appendix A, B, C
- [2] Makhoul, John. "Linear prediction: A tutorial review." Proceedings of the IEEE 63.4 (1975): 561-580.

Thank you